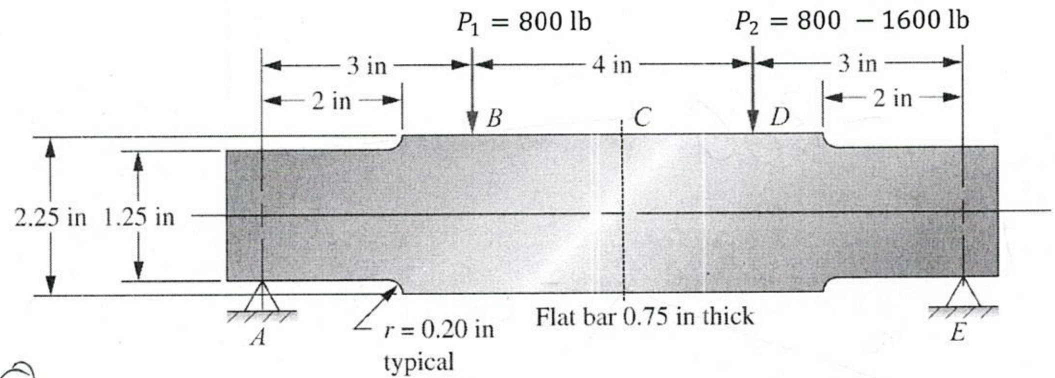
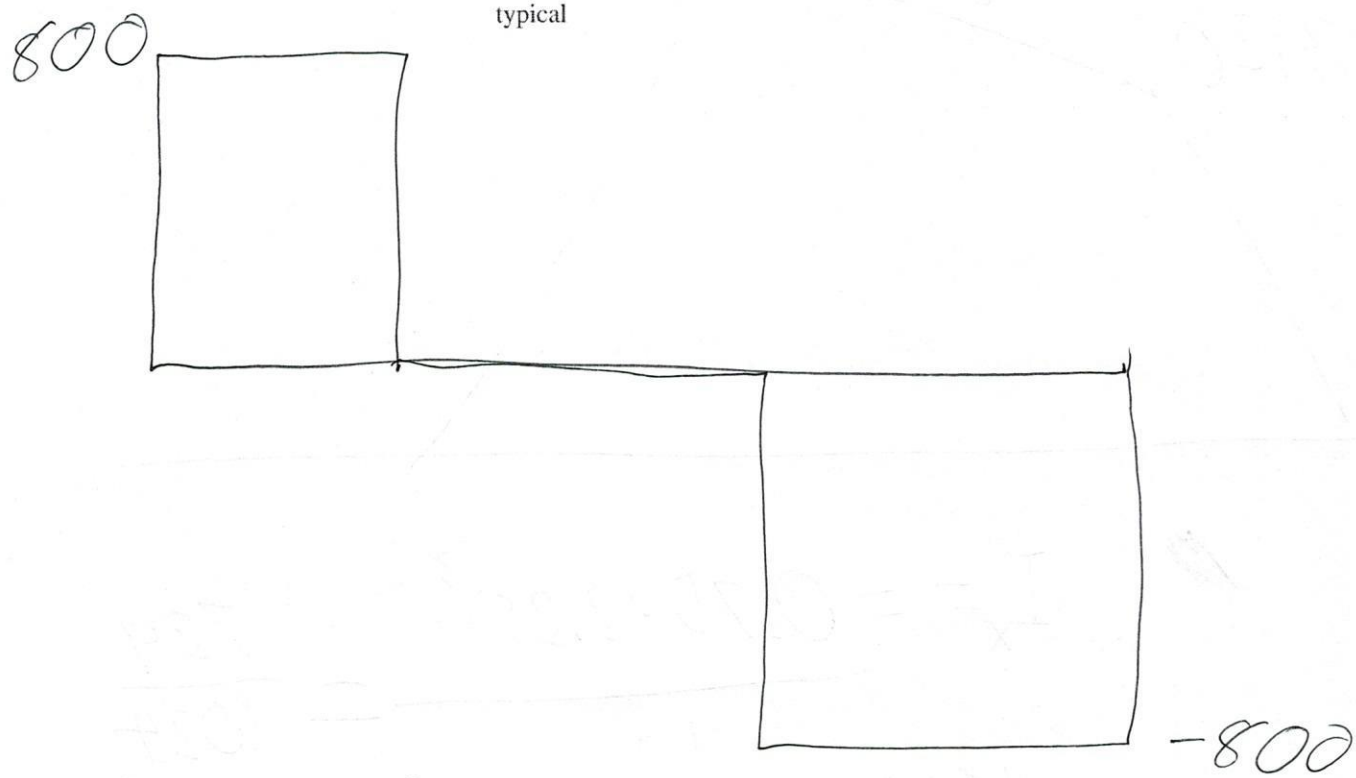
Machine Design (EMCH327 Section 001, Spring 2022)

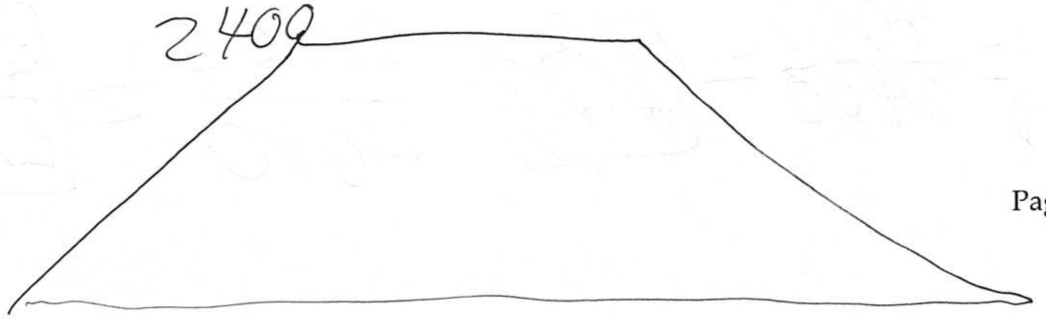
Mid-Term Exam 2, *8:30 AM, Wednesday, Mar. 16th, 2022*

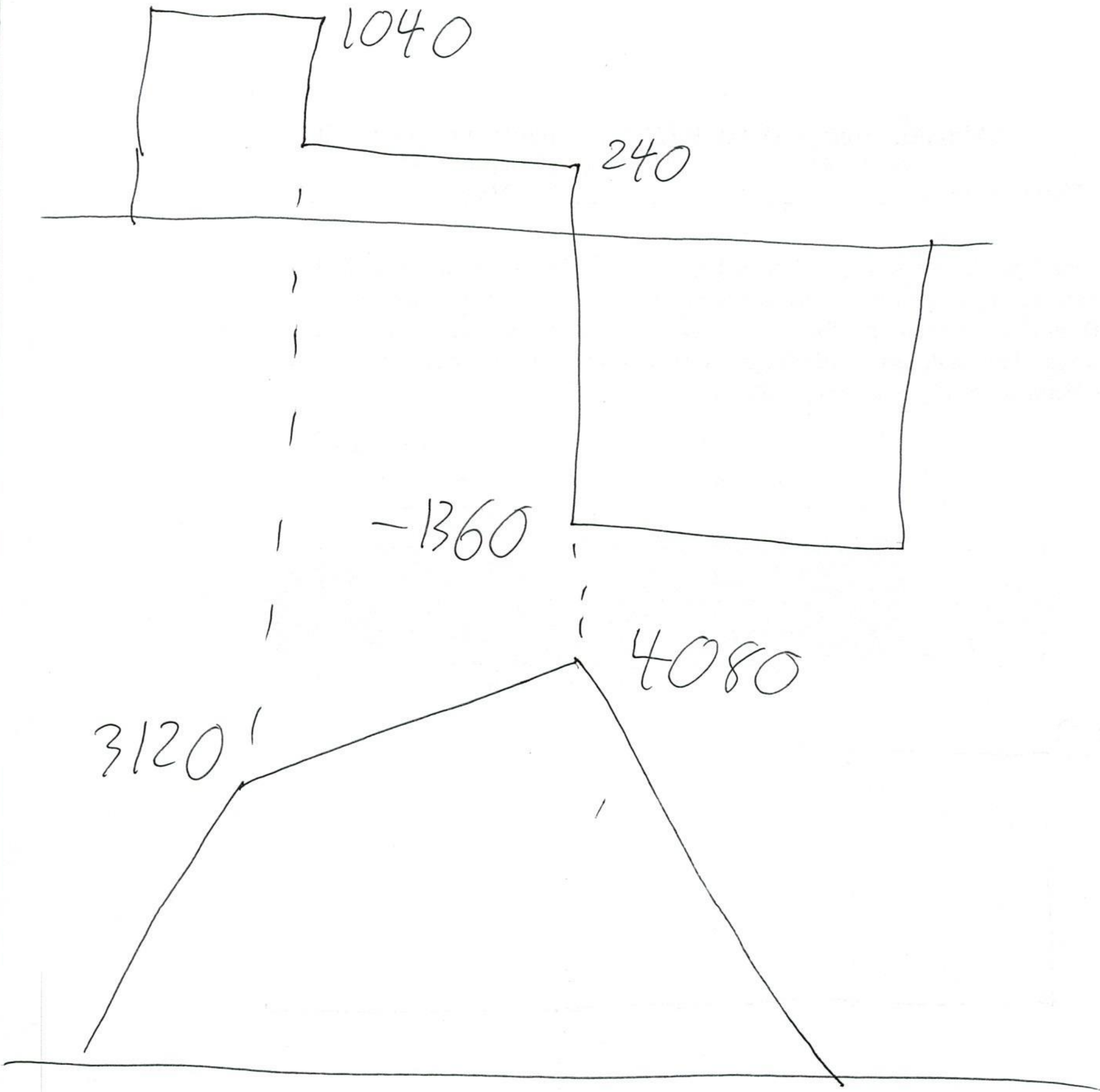
Name (First/Last): ______________________________ **Score:** ______________/100'

1. For a beam shown below, the left load $P_1 = 800$ lb remains constantly applied, while the right load P_2 is applied and changes from 800 lb to 1600 lb as the machine cycles. Determine the stress ratio **for the point *D***. Also, for the same point, use the maximum shear stress theory to determine the design factor so that the aluminium alloy 5052-H34 can be safely selected as material for the beam. (**40'** in total)









$$I_x = = \frac{0.75 \cdot (2.25)^3}{12} = \frac{729}{1024}$$

$$R_D = \cancel{\frac{4080}{2400}} = \frac{2406}{4080} = \boxed{\frac{10}{17}}$$

$$\sigma_{min} = \frac{2400\ lb\cdot in}{\left(\frac{729}{1024}\right)} = \frac{819200}{243}\ psi$$

$$\sigma_{max} = \frac{4080\ lb\cdot in}{\left(\frac{729}{1024}\right)} = 5731.028807\ psi$$

$$\sigma_1 = \sigma_{max} = 5731.028807\ psi$$

$$\tau^{abs}_{max} = \frac{\sigma_1}{2} = 2865.514404\ psi$$

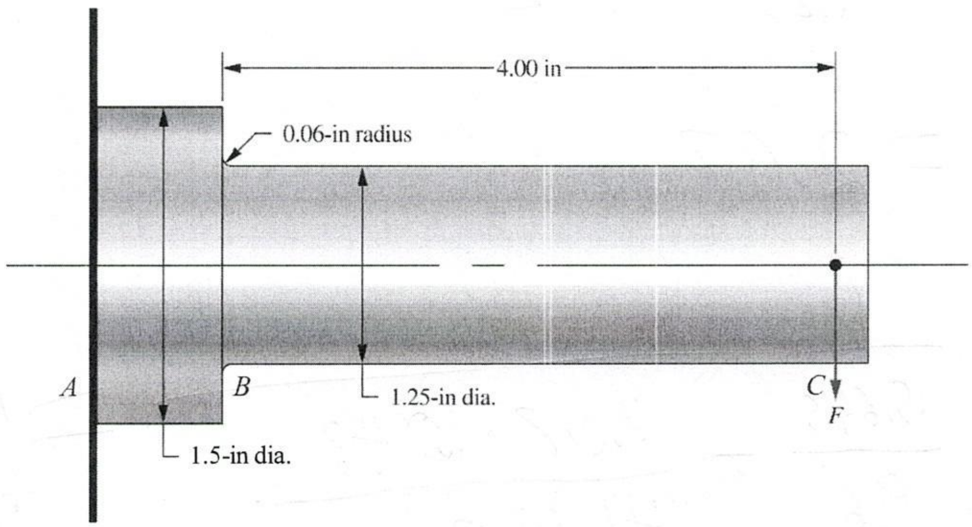
$$S_y = 31\ ksi$$

$$S_u = 38\ ksi$$

$$S_y = 31 = 2 \cdot \tau^{abs}_{max} \cdot N = 2 \cdot 2865.5 \cdot N = 31\ ksi$$

$$N = \boxed{5.4}$$

2. Use Goodman criterion to determine the design factor for a shaft with circular cross-section to carry a downward load that varies from 500 to 1000 lb. The material is wrought steel with $s_u = 96$ ksi and $s_n = 48$ ksi. Take reliability as 90%. <u>Stress concentration needs to be considered at point *B*.</u> (**30'** in total)



$$F_a = 500 \Rightarrow \sigma_a = \frac{4 \cdot 500}{\left(\frac{\pi \cdot 1.25^3}{32}\right)} = 10430.37885 \text{ psi}$$

$$F_m = 750 \Rightarrow \sigma_m = \frac{4 \cdot 750}{\left(\frac{\pi\, 1.25^3}{32}\right)} = 15645.56753 \text{psi}$$

$C_m = 1$

$C_{st} = 1$

$C_R = 0.9$

$$s_n' = 48 \cdot 1 \cdot 1 \cdot 0.9 \cdot 0.9535 = 41.1912$$

$$D_e = 0.37 \cdot 1.25 = 0.4625$$

$$C_s = \left(\frac{0.4625}{0.3}\right)^{-0.11} = 0.9535$$

$$r/d = \frac{0.06}{1.25} = 0.048$$

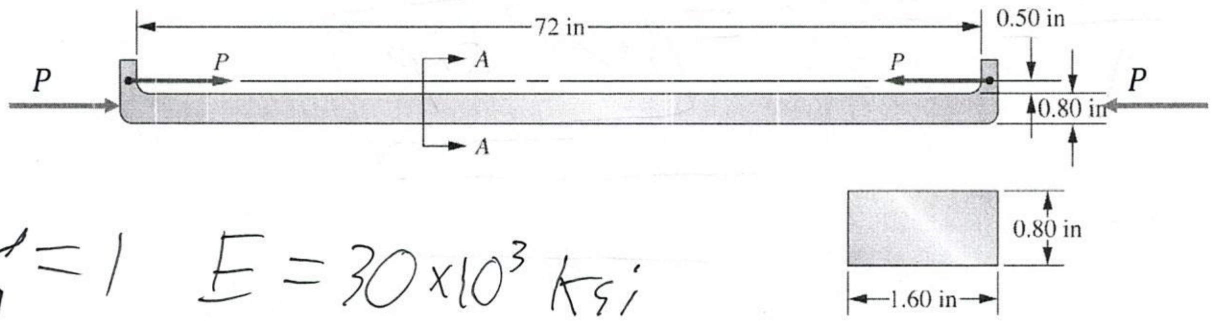
$$D/d = \frac{1.5}{1.25} = 1.2$$

$$Kt = 2.07$$

$$N = \frac{1}{\frac{15.645}{96} + \frac{2.07 \cdot 10.43}{41.1912}} = \boxed{1.455}$$

$S_y = 50$ $S_u = 89$

3. The column shown is subjected to opposing forces. The material is steel SAE 1117 WQT 350. <u>First</u>, determine the maximum allowable **centrical** load to achieve a design factor of 3. <u>Second</u>, determine whether the column is still safe when the same allowable load is **eccentrically** applied. (**30'** in total)



$K = 1$ $E = 30 \times 10^3$ ksi

$\cancel{P_{cr} = \frac{\pi^2 E I}{(KL)^2}}$

$I_x = \frac{1.6 \times 0.8^3}{12} = \left(\frac{128}{1875}\right)$

$\cancel{\frac{\pi^2 \cdot 30 \cdot 10^3 \cdot \left(\frac{128}{1875}\right)}{(1 \cdot 72)^2}} =$

$S_r = \frac{KL}{R = \sqrt{I/A}} = \frac{72}{\cancel{\quad}} = \frac{72}{(0.8/\sqrt{12})} = 311.769$

$C_c = \underline{110}$

$S_r > C_c$, so long column

$$P_{cr} = \frac{\pi^2 E I}{(L_e)^2} = \frac{\pi^2 \cdot 30 \cdot 10^3 \cdot 0.8 \cdot 1.6}{(311.769)^2} = 3.899 \text{ kip}$$

$$P_a = \frac{P_{cr}}{N} = \frac{3.899}{3} = \boxed{1.3 \text{ kip}}$$

Eccentrical $\cancel{\ }$ $e = 0.9$

$$r = \frac{0.8}{\sqrt{12}}$$

$$A = 0.8 \times 1.6 \quad c = \frac{1.6}{2} = 0.8$$

$$S_y = \frac{3 \cdot 1.3}{0.8 \cdot 1.6}\left(1 + \frac{0.9 \cdot 0.8}{\left(\frac{0.8}{\sqrt{12}}\right)^2} \cdot \sec\left(\frac{1.72}{2 \cdot \left(\frac{0.8}{\sqrt{12}}\right)}\sqrt{\frac{3 \cdot 1.3}{0.8 \cdot 1.6 \cdot 30 \times 10^3}}\right)\right.$$

$$\sec\left(\frac{72}{2 \cdot \frac{0.8}{\sqrt{12}}}\sqrt{\frac{3 \times 1.3}{0.8 \cdot 1.6 \cdot 30 \cdot 10^3}}\right)$$

[***Bonus Problem***] Using the boundary-lubricated approach, for a plain surface bearing to carry a radial load of 100 lb from a shaft having a diameter of 0.80 in and rotating at 800 rpm, determine the **design value** of pV factor. Use an L/D ratio of the bearing to be 1.25. (**10'** in total)

$L = 1.25 \times D = 1$

$P = \frac{100}{0.8} = 125$

$V = \frac{\pi \cdot 0.8 \cdot 800}{12} = 167.55$

$pV = 125 \cdot 167.55 = \boxed{20943.95}$ psi fpm

$2 \cdot 20943.95 = \boxed{41.885}$

Porous Bronze oil impregnated